

```
1 import java.util.*;
2 class Pendulum
3 {
4     static int arr[],size;
5
6     Pendulum(int a[],int sz)
7     {
8         size=sz;
9         arr=a;
10    }
11    void bubbleSort(int arr[])
12    {
13        int n = arr.length;
14        for (int i = 0; i < n-1; i++)
15            for (int j = 0; j < n-i-1; j++)
16                if (arr[j] > arr[j+1])
17                {
18                    // swap arr[j+1] and arr[j]
19
20                    int temp = arr[j];
21                    arr[j] = arr[j+1];
22                    arr[j+1] = temp;
23                }
24    }
25
26    Pendulum PendulumSort(int arr[], int n)
27    {
28        Pendulum p=new Pendulum(arr,size);
29        // sorting the elements
30        bubbleSort(arr);
31        // Auxiliary array to store output
32        int op[] = new int[n];
33        // calculating the middle index
34        int mid = (n-1)/2;
35        // storing the minimum element in the middle
36        // i is index for output array and j is for
37        // input array.
38        int j = 1, i = 1;
39        p.arr[mid] = arr[0];
40        for (i = 1; i <= mid; i++)
41        {
42            p.arr[mid+i] = arr[j++];
43            p.arr[mid-i] = arr[j++];
44        }
45
46        // adjustment for when no. of elements is even
47        if (n%2 == 0)
48            p.arr[mid+i] = arr[j];
49        return p;
50    }
```

```
51     void display()
52     {
53         // Printing the pendulum arrangement
54         System.out.println("Pendulum arrangement:");
55         for (int i = 0 ; i < size; i++)
56             System.out.print(arr[i] + " ");
57         System.out.println();
58     }
59
60     // Driver method
61     public static void main()
62     {
63         Pendulum ob=new Pendulum(arr,size);
64         Pendulum obj=obj;
65         obj=obj.PendulumSort(arr, arr.length);
66         obj.display();
67     }
68 }
```